Numerical Weather Prediction (NWP)

MATH406 Individual Project: Multi-Variable Temperature Forecasting with ERA5 Data

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Research Question

Core Objective

Can statistical models informed by physical diagnostics provide accurate short-term temperature forecasts for East Asian cities?

Scientific Goals:

- Quantify forecast skill degradation with lead time (1h, 6h, 12h)
- Identify geographical patterns in predictability
- Assess contributions of physical terms to predictive performance
- Compare regularization strategies in regression modeling

Mathematical Framework

Core Regression Model

$$T(t + \Delta t) = \beta_0 + \sum_{i=1}^{p} \beta_i X_i(t) + \epsilon$$

where X_i includes:

- Lagged temperature: T_{t-1}, T_{t-2}
- Meteorological variables: u_{10}, v_{10}, p_{msl}
- Physical diagnostics: $dT/dt, u\nabla T + v\nabla T$

Mathematical Framework

Regularization Techniques

$$\hat{\beta} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|^q \right\}$$

- Ridge ($q = 2$): L2 penalty, shrinks coefficients
- Lasso ($q = 1$): L1 penalty, performs feature selection

Data & Methodology

Data Source:

- ERA5 Reanalysis (ECMWF)
- July 1-4, 2025, hourly data
- Variables:
  2m temperature, 10m winds, mean sea level pressure

Target Cities:

- Shanghai, Suzhou, Guangzhou, Hong Kong, Lanzhou
- Seoul, Busan, Fukuoka

Model Pipeline:

1. Data extraction at city coordinates
2. Feature engineering
3. Train/test split (70/30)
4. Hyperparameter tuning (GridSearchCV)
5. Multi-step forecasting (1h, 6h, 12h)
6. Performance evaluation (MAE, RMSE)

Forecast Skill Degradation

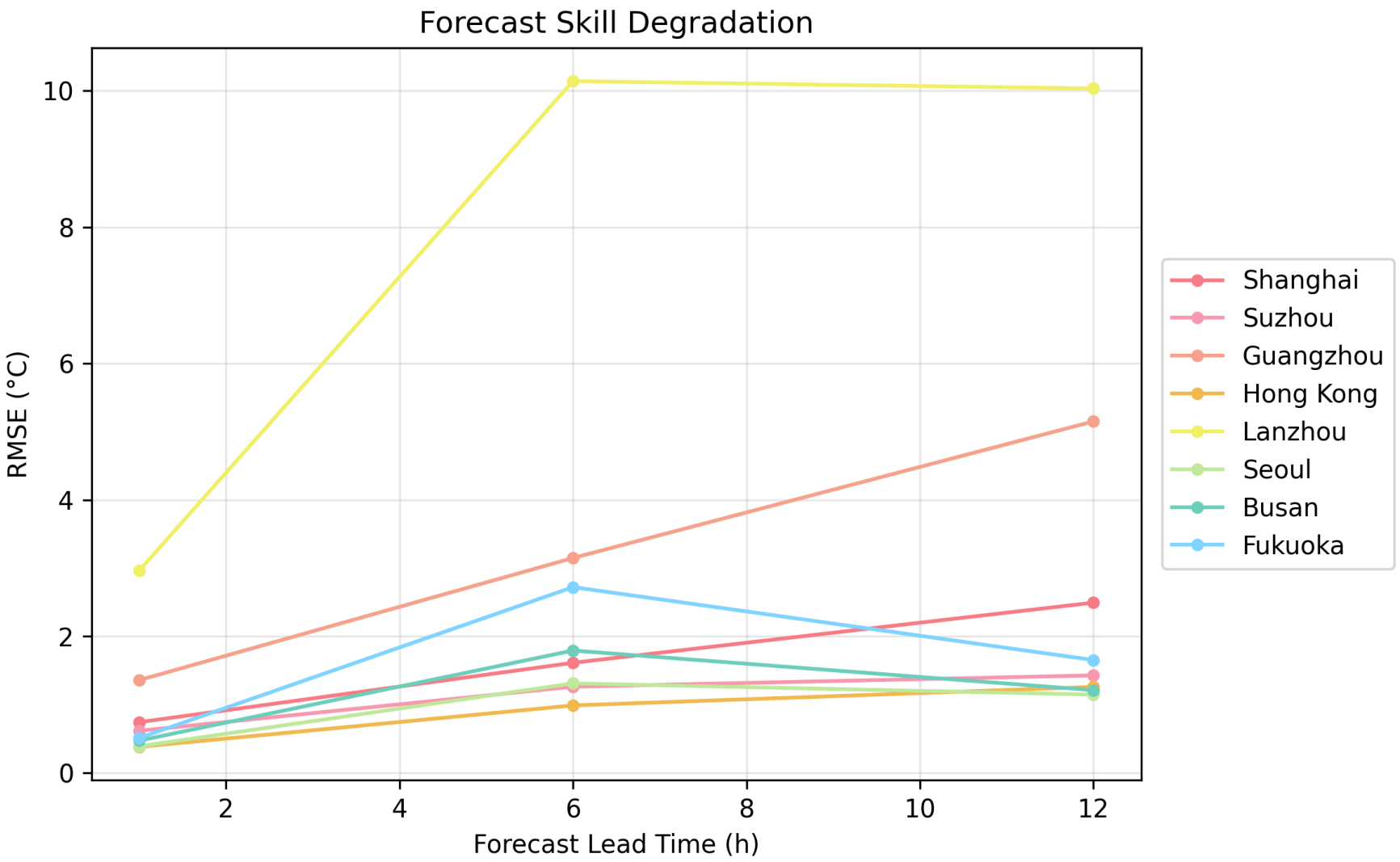

Figure: RMSE increases systematically with lead time for all cities

City Comparison at 6h Forecast

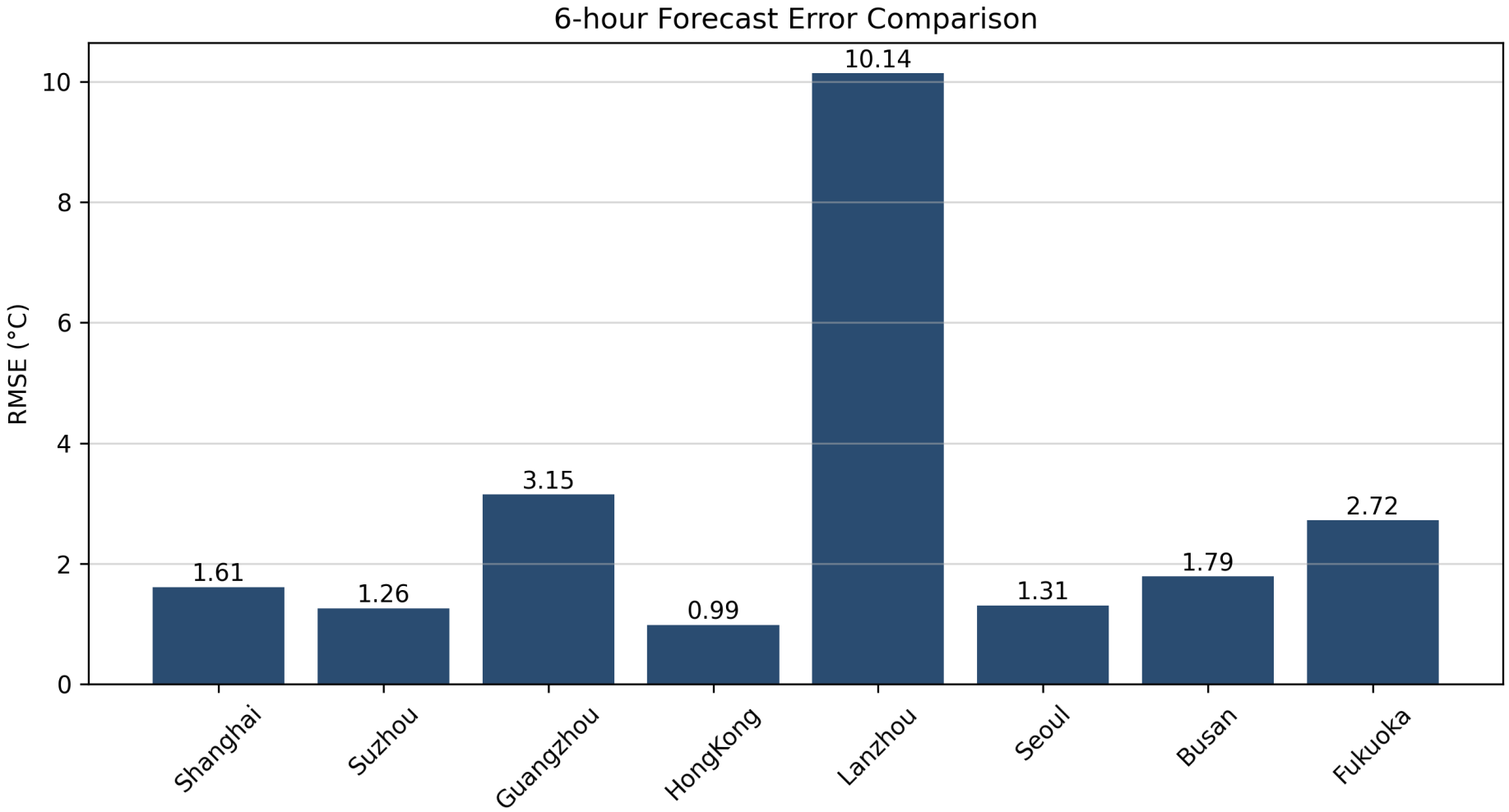

Figure: Coastal cities show lower RMSE than inland locations

Physical Diagnostics: Shanghai Example

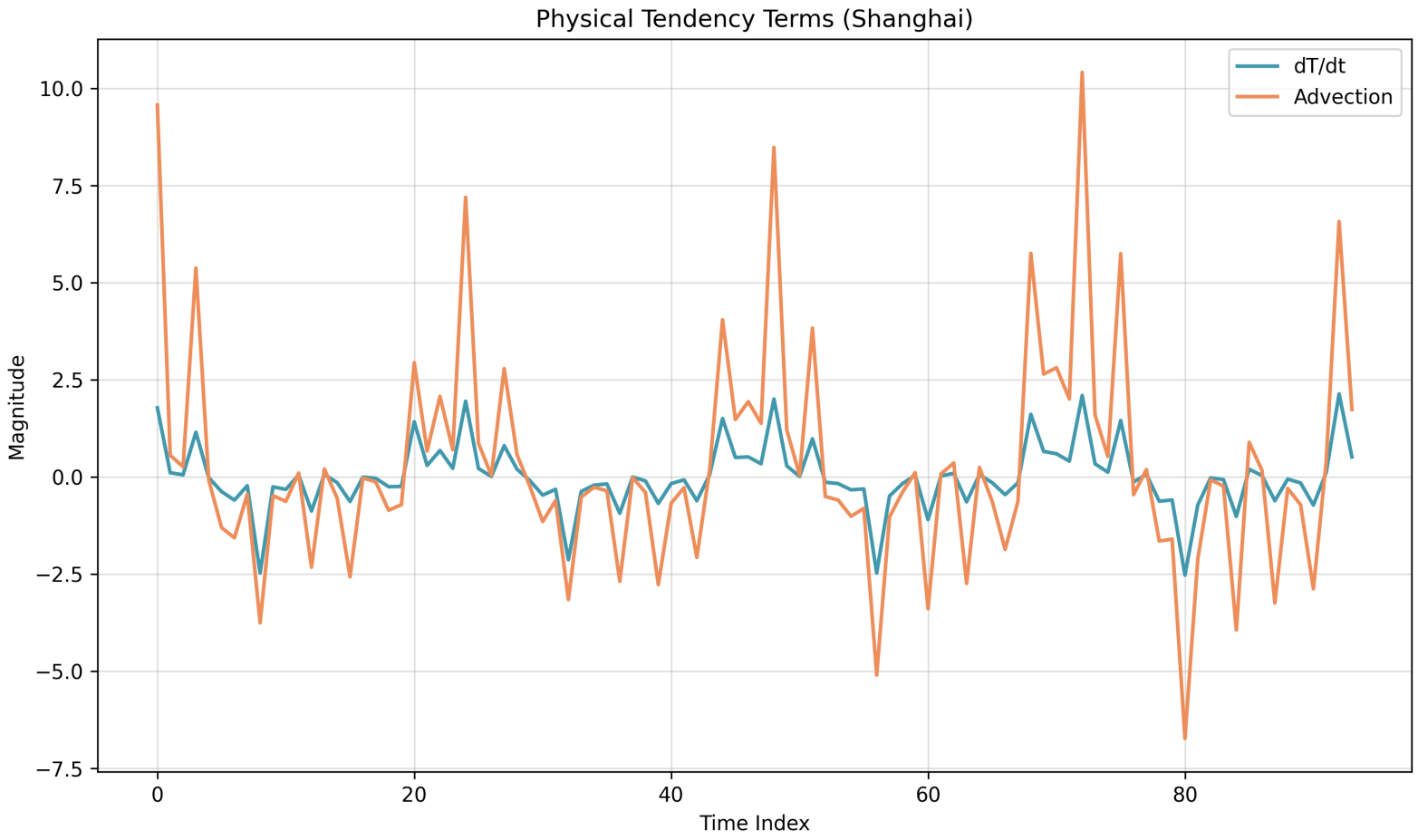

Figure: Temperature tendency and horizontal advection terms

Physical terms capture dynamic processes and enhance interpretability.

Forecast vs Observation (Shanghai, 1h lead)

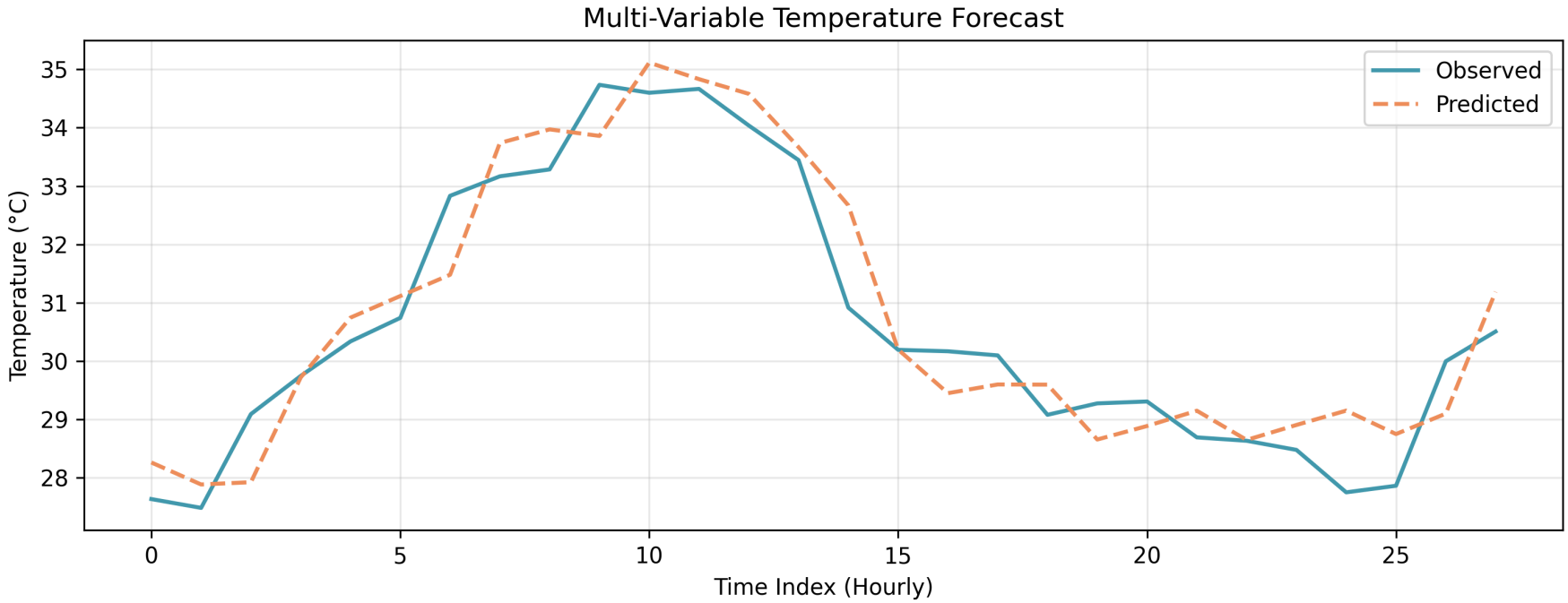

Figure: Ridge regression predictions vs actual observations

Metric	Value
MAE	0.616 °C
RMSE	0.741 °C

Model Evaluation Results

Ridge Regression with GridSearchCV

Hyperparameter	Value
Best Alpha	0.234
MAE	0.616 °C
RMSE	0.741 °C
Training Split	70%
Testing Split	30%

- Ridge (fixed $\alpha = 1.0$): MAE $= 0.550\,°\mathrm{C}$, $RMSE = 0.707\,°\mathrm{C}$
- Lasso (fixed $\alpha = 1.0$): MAE $= 0.551\,°\mathrm{C}$, $RMSE = 0.707\,°\mathrm{C}$
- Ridge (auto-tuned): MAE $= 0.553\,°\mathrm{C}$, $RMSE = 0.706\,°\mathrm{C}$
- Lasso (auto-tuned): MAE $= 0.543\,°\mathrm{C}$, $RMSE = 0.713\,°\mathrm{C}$

Regression Coefficients Analysis (Shanghai, 1h forecast)

Variable	Coefficient	Physical Meaning
T_lag1	0.996	Temperature persistence
dT/dt	0.677	Local temperature tendency
U_lag1	0.491	Zonal wind advection
V_lag1	-0.198	Meridional wind effect
P_lag1	0.073	Pressure adjustment
Adv_T	0.080	Horizontal advection
T_lag2	-0.115	Second-order memory

Key Insights:

- Temperature persistence dominates (coefficient: 0.996)
- Physical terms (dTdt, Adv_T) contribute positively
- Model provides physically interpretable relationships

Results & Findings

Key Findings

1. **Predictability decay**: RMSE increases systematically with lead time (1h→12h)
2. **Geographical pattern**: Coastal cities show 2–11× lower RMSE than inland (Lanzhou)
3. **Model performance**: Ridge regression achieves competitive skill with 0.74°C RMSE for 1h forecasts
4. **Physical interpretability**: Diagnostic terms enhance model understanding
5. **Feature importance**: Temperature persistence dominates short-term forecasts

Limitations & Future Work

Methodological Limitations

- **Linear assumptions**: Cannot capture nonlinear atmospheric dynamics
- **Stationarity**: July-only data may not generalize to other seasons
- **Spatial independence**: City-by-city approach ignores spatial correlations
- **Simplified physics**: Missing radiation, convection, and boundary layer processes

Future Directions

- Incorporate neural networks for nonlinear relationships
- Extend to seasonal analysis and extreme weather events
- Include spatial dependencies via graph neural networks
- Integrate more complete physical parameterizations

Towards Full Numerical Weather Modeling

This project adopts a physics-informed statistical approach. A natural next step is to solve governing atmospheric PDEs using numerical methods.

Proposed PDE System

Future work will implement simplified dynamical cores based on:

- Primitive Equations (Momentum, Thermodynamics, Continuity)
- Hydrostatic and Geostrophic Approximations
- Advection-Diffusion Equation for Temperature

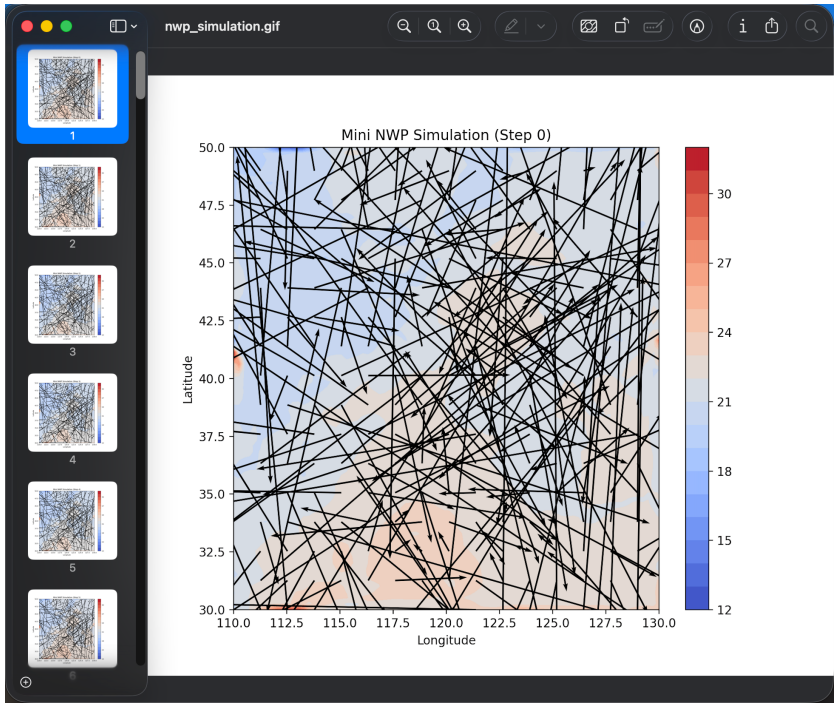

Figure: Prototype numerical simulation initialized from ERA5 data, illustrating coupled temperature and wind evolution.

Project Milestones

Week	Main Tasks	Deliverables
Week 6	Model improvement, feature refinement, and development of numerical PDE solver; testing numerical stability and parameter sensitivity	Improved regression models; preliminary PDE simulation results
Week 7	Final report writing, code and data organization, figure generation, and reproducibility checks	Complete draft of report; organized dataset and code repository
Week 8	Final polishing of report and slides; improvement of PDE simulations and visualization	Final report and slides; refined numerical simulation results